

## NORTH BAY, ON, Canada

WMO: 717310

Lat: 46.365N

Lon: 79.423W

Elev: 370

StdP: 96.95

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -27.7      | -24.7 | -32.5                         | 0.2 | -27.2 | -29.6 | 0.3 | -24.4 | 11.4                  | -8.0 | 10.3 | -8.4 | 2.9                   | 340  | 0.632 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.4                    | 28.0            | 20.1 | 26.4 | 19.2 | 25.0 | 18.4 | 21.9                | 25.8 | 20.8 | 24.1 | 19.9 | 23.0 | 5.0                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.6                            | 16.0 | 23.5 | 19.7 | 15.1 | 22.6 | 18.7 | 14.2 | 21.8 | 65.9          | 25.9 | 62.0 | 24.2 | 58.7 | 23.0 | 26.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.1              | 8.9  | 8.0 | -32.1                      | 30.8 | 3.7                | 1.5 | -34.8                                              | 31.8 | -36.9      | 32.7 | -39.0      | 33.5 | -41.7      | 34.5 |
|                   |      |     | -32.3                      | 23.8 | 3.6                | 1.2 | -34.9                                              | 24.7 | -37.0      | 25.4 | -39.1      | 26.1 | -41.7      | 27.0 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 4.6         | -12.1 | -10.2 | -4.4  | 3.4   | 11.5  | 16.7  | 19.0  | 18.0  | 13.9  | 6.9   | -0.5  | -7.5  | (6)  |
|                                                                        | DBStd       | 12.17       | 7.74  | 6.62  | 6.72  | 5.31  | 4.90  | 3.94  | 3.06  | 3.18  | 4.38  | 4.92  | 5.59  | 6.60  | (7)  |
|                                                                        | HDD10.0     | 2944        | 686   | 567   | 446   | 206   | 41    | 2     | 0     | 0     | 14    | 124   | 316   | 542   | (8)  |
|                                                                        | HDD18.3     | 5126        | 944   | 800   | 703   | 447   | 217   | 75    | 29    | 45    | 145   | 356   | 565   | 800   | (9)  |
|                                                                        | CDD10.0     | 991         | 0     | 0     | 1     | 9     | 89    | 205   | 279   | 249   | 130   | 27    | 1     | 0     | (10) |
|                                                                        | CDD18.3     | 131         | 0     | 0     | 0     | 0     | 6     | 28    | 51    | 35    | 11    | 1     | 0     | 0     | (11) |
|                                                                        | CDH23.3     | 727         | 0     | 0     | 0     | 2     | 46    | 171   | 281   | 178   | 47    | 2     | 0     | 0     | (12) |
|                                                                        | CDH26.7     | 118         | 0     | 0     | 0     | 0     | 7     | 29    | 51    | 27    | 4     | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 3.9         | 4.0   | 4.1   | 4.2   | 4.4   | 4.0   | 3.6   | 3.5   | 3.4   | 3.5   | 4.0   | 4.1   | 4.0   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 1012        | 67    | 56    | 64    | 68    | 85    | 93    | 101   | 101   | 115   | 94    | 92    | 79    | (15) |
|                                                                        | PrecMax     | 1257        | 151   | 108   | 127   | 125   | 180   | 160   | 256   | 234   | 191   | 156   | 168   | 160   | (16) |
|                                                                        | PrecMin     | 770         | 26    | 13    | 12    | 21    | 17    | 22    | 16    | 26    | 48    | 15    | 15    | 42    | (17) |
|                                                                        | PrecStd     | 121         | 24    | 24    | 31    | 25    | 39    | 38    | 54    | 43    | 40    | 34    | 34    | 26    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 7.4   | 6.8   | 15.0  | 21.8  | 27.6  | 29.4  | 30.5  | 29.5  | 27.1  | 22.5  | 14.8  | 8.2   | (19) |
|                                                                        |             | MCWB        | 7.1   | 5.3   | 10.5  | 13.2  | 18.6  | 20.8  | 21.8  | 21.2  | 19.8  | 16.4  | 10.9  | 7.6   | (20) |
|                                                                        | 2%          | DB          | 3.0   | 3.6   | 10.1  | 17.5  | 24.3  | 27.3  | 28.0  | 27.1  | 24.6  | 19.2  | 11.6  | 4.8   | (21) |
|                                                                        |             | MCWB        | 2.5   | 1.9   | 6.5   | 10.0  | 15.6  | 19.5  | 20.2  | 20.1  | 19.1  | 15.1  | 9.5   | 4.1   | (22) |
|                                                                        | 5%          | DB          | 0.6   | 1.1   | 7.0   | 14.2  | 22.0  | 25.3  | 26.2  | 25.3  | 22.5  | 16.7  | 9.3   | 2.7   | (23) |
|                                                                        |             | MCWB        | -0.1  | -0.2  | 4.0   | 8.4   | 14.5  | 18.3  | 19.3  | 18.8  | 17.9  | 13.4  | 7.7   | 2.1   | (24) |
|                                                                        | 10%         | DB          | -1.1  | -0.6  | 4.6   | 11.4  | 19.6  | 23.4  | 24.7  | 23.6  | 20.6  | 14.3  | 7.1   | 1.0   | (25) |
|                                                                        |             | MCWB        | -1.8  | -1.5  | 1.9   | 6.1   | 13.3  | 17.1  | 18.5  | 18.2  | 17.0  | 11.8  | 5.7   | 0.4   | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 7.0   | 5.8   | 11.7  | 14.8  | 20.1  | 22.7  | 23.5  | 22.8  | 21.6  | 17.9  | 12.2  | 8.0   | (27) |
|                                                                        |             | MCDB        | 7.2   | 6.7   | 13.7  | 19.7  | 25.5  | 27.5  | 28.0  | 26.6  | 24.8  | 20.8  | 13.4  | 8.2   | (28) |
|                                                                        | 2%          | WB          | 2.5   | 2.1   | 6.7   | 12.0  | 17.7  | 21.1  | 22.0  | 21.3  | 20.2  | 15.6  | 9.9   | 4.2   | (29) |
|                                                                        |             | MCDB        | 2.9   | 3.4   | 9.1   | 15.1  | 21.9  | 25.3  | 25.8  | 24.7  | 22.5  | 18.5  | 11.3  | 4.5   | (30) |
|                                                                        | 5%          | WB          | 0.0   | -0.1  | 4.3   | 9.1   | 16.0  | 19.7  | 20.8  | 20.2  | 18.9  | 13.8  | 7.8   | 2.2   | (31) |
|                                                                        |             | MCDB        | 0.4   | 0.8   | 6.5   | 13.3  | 20.1  | 23.1  | 24.0  | 23.4  | 21.7  | 16.1  | 9.0   | 2.8   | (32) |
|                                                                        | 10%         | WB          | -1.8  | -1.7  | 2.0   | 6.8   | 14.3  | 18.4  | 19.8  | 19.2  | 17.5  | 12.0  | 5.7   | 0.5   | (33) |
|                                                                        |             | MCDB        | -1.1  | -0.6  | 4.3   | 10.7  | 18.3  | 21.7  | 23.0  | 22.2  | 20.1  | 14.2  | 7.0   | 0.9   | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.9   | 9.2   | 9.3   | 9.5   | 10.2  | 9.7   | 9.4   | 9.1   | 9.0   | 7.5   | 6.7   | 7.0   | (35) |
|                                                                        |             | MCDBR       | 7.9   | 9.1   | 10.3  | 13.2  | 12.8  | 11.3  | 11.0  | 10.7  | 10.5  | 10.6  | 9.6   | 7.2   | (36) |
|                                                                        | 5% WB       | MCWBR       | 7.9   | 8.3   | 7.4   | 8.4   | 6.9   | 5.7   | 5.1   | 5.2   | 5.9   | 6.9   | 8.1   | 7.1   | (37) |
|                                                                        |             | MCWBR       | 7.9   | 8.8   | 9.5   | 11.7  | 10.5  | 9.5   | 9.3   | 8.8   | 9.2   | 9.8   | 9.0   | 7.1   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.265 | 0.279 | 0.310 | 0.346 | 0.356 | 0.387 | 0.389 | 0.381 | 0.359 | 0.330 | 0.295 | 0.267 | (40) |
|                                                                        |             | taud        | 2.349 | 2.292 | 2.323 | 2.369 | 2.413 | 2.348 | 2.370 | 2.387 | 2.434 | 2.504 | 2.527 | 2.418 | (41) |
|                                                                        | Ebn at Noon |             | 840   | 898   | 917   | 910   | 909   | 878   | 872   | 865   | 855   | 833   | 807   | 801   | (42) |
|                                                                        |             | Edh at Noon | 73    | 95    | 108   | 113   | 113   | 121   | 117   | 111   | 97    | 78    | 63    | 61    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.29  | 2.21  | 3.65  | 4.52  | 5.20  | 5.65  | 5.63  | 4.83  | 3.68  | 2.20  | 1.21  | 0.96  | (44) |
|                                                                        | RadStd      |             | 0.18  | 0.28  | 0.37  | 0.37  | 0.46  | 0.42  | 0.42  | 0.30  | 0.39  | 0.26  | 0.14  | 0.09  | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (47) | Regional (15 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page